

Vitor Gama

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PROFESSIONAL SUMMARY

Innovative and detail-oriented Chemical Engineer with expertise in process systems engineering, with focus on process simulation and optimization, process operability analysis and modeling. Adept at applying cutting-edge research and technical solutions to optimize efficiency, quality, and integrity in chemical processes. Passionate about leveraging ingenuity and accountability to solve complex engineering challenges while maintaining strong leadership and collaborative skills.

KEY SKILLS

- **Process Optimization & Efficiency:** Developed process models and simulation frameworks to enhance membrane DAC system performance.
- **Technical Expertise & Ingenuity:** Extensive experience with AVEVA Process simulation, Aspen Plus, Aspen Custom Modeler, HYSYS, and Python for process simulation and data analysis.
- **Safety & Compliance:** Adhered to best practices in chemical process safety and environmental regulations.
- **Software Development & Automation:** Designed cost estimation tool using C# and XAML for chemical process simulations.

EDUCATION

West Virginia University

Ph.D. student, Chemical Engineering

GPA: 3.75/4.00

Morgantown, WV

Aug. 2021 – Dec. 2025 (Expected)

Federal University of Campina Grande

M.Sc., Chemical Engineering

GPA: 9.14/10.00

Campina Grande, Paraiba, Brazil

Sept. 2018 – April 2021

Federal University of Campina Grande

B.Sc., Chemical Engineering

GPA: 8.18/10.00

Campina Grande, Paraiba, Brazil

May 2013 – Aug. 2018

AWARDS & ACHIEVEMENTS

- AIChE Environmental Division Graduate Paper Award (2024) – 2nd place for research on membrane DAC process operability.
- Jack and Marietta Mullenger Fellowship (2024) – Recognized for academic excellence and research contributions.
- Statler College PhD Recruitment Fellowship (2021).

RESEARCH EXPERIENCE

West Virginia University

Graduate Research Assistant (Ph.D.)

Morgantown, WV

Aug. 2021 – Present

- Developed simulation and operability analysis frameworks for gas separation membranes in Direct Air Capture (DAC) with Carbon Capture Utilization and Storage (CCUS).
- Applied inverse design and process operability analysis to optimize system performance.
- Published in peer-reviewed journals, demonstrating commitment to quality and research integrity.

Federal University of Campina Grande

Graduate Research Assistant (M.Sc.)

Campina Grande, Paraiba, Brazil

Sept. 2018 – April 2021

- Designed and implemented **CostApp**, a cost estimation tool improving engineering decision-making efficiency.
- Developed process control routines in collaboration project with Petrobras, focusing on automation and efficiency enhancements.

TECHNICAL PROFICIENCIES

- **Programming:** Python, MATLAB, C#, JavaScript, LaTeX.
- **Process Simulation:** Aspen Plus, HYSYS, PRO/II, AVEVA Process Simulation, ChemCad.
- **Version Control:** Git, GitHub, GitLab.
- **Languages:** English, Portuguese.

PROFESSIONAL LEADERSHIP & SERVICE

- President, Brazilian Student Association (2022 – 2024).
- Member, Statler College DEI Student Committee (2022 – 2023).
- Research Group Leader, CODES (2022 – Present).

CLIFTONSTRENGTHS

Arranger | Positivity | Input | Woo | Communication